Siwei Meng

Ph.D. Candidate, Computer Science and Engineering, University of Nevada, Reno
swee.swm@gmail.com | +1-775-338-2335 | Personal Site | GitHub | Google Scholar

Research Interests

Physical AI for generative world models: diagnosing whether generated video obeys physics, using physical plausibility as a training signal, and grounding generation in simulation. Four first-author papers at top-tier venues on this line in two years (NeurIPS 2026, ECCV 2026, ACM MM 2026 Oral, IJCAI 2025), spanning benchmark, post-training, simulation-guided generation, and a survey of physical priors in generative AI. Long-term goal: world models whose physical behavior can be measured, verified, and improved.

Education

University of Nevada, Reno **Aug 2024 – Present**

Ph.D. in Computer Science and Engineering (advisor: Prof. Ping Liu). GPA 3.9/4.0. Graduate Dean's Merit Scholarship.

Shenzhen University (U.S. News Best Global Universities for Computer Science, top 80) **2021 – 2024**

M.E. in New Generation Electronic Information Technology. Special Freshman Scholarship. Focus: computer vision.

South China University of Technology (U.S. News Best Global Universities for Computer Science, top 80) **2017 – 2021**

B.Eng. in Information and Interaction Design. School Second Class Scholarship.

Publications

First-author papers are marked ⋆. Full venue names; all conference papers are peer-reviewed main-track publications unless noted.

⋆ **Siwei Meng**, Xuerou Cai, Rui Hu, Ping Liu. **PhyMetric: Diagnosing Physical Plausibility in Text-to-Video Generation via Scene-Level QA**. *Conference on Neural Information Processing Systems (NeurIPS 2026), Datasets and Benchmarks Track.*

⋆ **Siwei Meng**, Yawei Luo, Ping Liu. **PhyMAGIC: Physical Motion-Aware Generative Inference with Confidence-Guided LLM**. *European Conference on Computer Vision (ECCV 2026).*

⋆ **Siwei Meng**, Yawei Luo, Shu Zhang, Ping Liu. **When Physical Preferences Meet Semantic Constraints: Physical and Semantic Direct Preference Optimization for Text-to-Video Generation**. *ACM International Conference on Multimedia (ACM MM 2026).* **Oral presentation**.

⋆ **Siwei Meng**, Yawei Luo, Ping Liu. **Grounding Creativity in Physics: A Brief Survey of Physical Priors in AIGC**. *International Joint Conference on Artificial Intelligence (IJCAI 2025), Survey Track.*

⋆ **Siwei Meng**, Wuzhen Shi, Yang Wen. **Occlusion-Aware Learning and Facial Action Unit Priors for Facial Expression Recognition**. *IEEE International Conference on Multimedia and Expo (ICME 2026).*

⋆ **Siwei Meng**, Wuzhen Shi. **Fusing Structure and Appearance Features in Facial Expression Recognition Transformer**. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2024).*

1. Yuqi Li, **Siwei Meng**, Chuanguang Yang, et al. **A Comprehensive Survey of Interaction Techniques in 3D Scene Generation**. *International Joint Conference on Artificial Intelligence (IJCAI 2026), Survey Track.*

Research Experience

University of Nevada, Reno **Graduate Research Assistant** **Sep 2024 – Present**

Physical AI for video generation and world models, advised by Prof. Ping Liu

- **Diagnosing physics in generated video (PhyMetric, NeurIPS 2026).** Led the design of a scene-level diagnostic benchmark of 1,115 physical scenes across mechanics, materials, chemistry, thermal phenomena, fluids, and electromagnetism. Replaced holistic plausibility scores with structured physical priors, question-aware frame sampling, and VLM-based binary judgments that localize specific failure modes. Evaluated open-source T2V models and VLM judges and showed that scene-specific QA aligns better with human judgment than holistic scoring.
- **Physics as a training signal (PSDPO, ACM MM 2026 Oral).** Built a 10K-pair physical preference dataset from multimodal physical evaluators, optical-flow and depth evidence, and ViCLIP semantic scores; formulated constrained preference learning with per-sample semantic feasibility, a staged training schedule, and semantic-aware gradient suppression so that improving physics does not degrade prompt fidelity. LoRA-based DPO on Wan-style video models.

- **Simulation-guided generation (PhyMAGIC, ECCV 2026).** Training-free pipeline for physically consistent dynamic 3D generation from one image and a text instruction: an image-to-video model generates targeted probe videos, a VLM estimates physical parameters with confidence, and a differentiable MPM simulator initialized from 3D Gaussian reconstruction produces the final motion. Improved physical plausibility and text-motion alignment over generative and simulation baselines.

Industry Experience

Meitu, California, USA — **Research Intern** — **Jul 2026 – Present**

Long-video temporal grounding and multimodal post-training

- Built a long-video temporal grounding benchmark of 194 videos and 2,268 event queries with a human–VLM annotation pipeline (MLLM event verification, boundary refinement, human review).
- Evaluated video-language models on fine-grained event localization and analyzed accuracy–efficiency trade-offs of frame sampling and sliding-window inference.

Parametrix.ai — **AIGC Algorithm Intern** — **May – Aug 2023**

- Trained and evaluated generative models for controllable anime facial expression synthesis; defined a hierarchical expression taxonomy used for data organization and prompt design.

TCL Co., Ltd. — **Graphic Semantic Algorithm Intern** — **Jul – Dec 2022**

- Developed lightweight YOLOv3 and MobileNet-based recognition models for short-video quality screening, including open/closed-eye detection for high-speed video.

Research Funding

Co-PI, FirePrune: Multi-modal Dataset and MLLM-Guided Model Pruning for AI-Powered Wildfire Detection **2025 – 2026**

NSF EPSCoR HDRFS Seed Grant, $30,000 (PI: Ping Liu). Model pruning and compression for reliable wildfire detection on resource-constrained devices.

Honors and Awards

Oral Presentation, ACM International Conference on Multimedia	2026
Graduate Dean's Merit Scholarship, University of Nevada, Reno	2024
Special Freshman Scholarship, Shenzhen University	2021
School Second Class Scholarship, South China University of Technology	2017 – 2021

Technical Skills

Physics-grounded generation: differentiable MPM simulation, 3D Gaussian Splatting, video diffusion models (Wan-style), CUDA, PyTorch.
Post-training and evaluation: DPO, GRPO, RLVR, LoRA fine-tuning, preference data construction, VLM-as-judge benchmarks for video generation.

Professional Service

Conference reviewer: NeurIPS 2026, AAAI 2027, IJCNN 2027, ICME 2026, ICME Workshops 2026, SIGIR 2026 Workshop.
Journal reviewer: IEEE Transactions on Multimedia, IEEE Open Journal of the Communications Society.